
FAIRFED+

Abstract

Federated learning (FL) enables privacy-preserving collaborative model training across distributed clients but raises significant fairness challenges when demographic data distributions are heterogeneous. Existing approaches either enforce fairness globally through re-weighted aggregation (FairFed) or locally through a fixed regularization penalty (EquiFL), but neither closes the feedback loop between server-level fairness signals and per-client training objectives. We propose FairFed+, a unified algorithm that adaptively scales each client’s local fairness regularization penalty in proportion to its deviation from the global fairness metric, while preserving FairFed’s fairness-aware server aggregation unchanged. This design introduces zero additional communication overhead and requires no new hyperparameters beyond the existing base penalty. Theoretical analysis shows that FairFed+ reduces the worst-client fairness gap more aggressively than either baseline, and we provide a full algorithmic specification, motivation, and discussion of expected outcomes and limitations.

1 Introduction and Motivation

Machine learning models are increasingly deployed in high-stakes decision-making domains such as healthcare, credit scoring, and recruitment. In these contexts, ensuring that a model does not discriminate against individuals based on sensitive demographic attributes such as gender, race, or age is not merely desirable but ethically and legally imperative. Fairness in machine learning has therefore emerged as a central research concern, with a growing body of work developing formal criteria, training algorithms, and post-processing techniques to mitigate algorithmic bias.

Federated learning (FL) has become an attractive paradigm for privacy-preserving model training. Instead of pooling raw data at a central server, FL allows a collection of clients to collaboratively train a shared global model by exchanging only model updates. The server aggregates these updates into a new global model, and the cycle repeats. This architecture naturally avoids direct data sharing, aligning with regulatory frameworks such as GDPR. However, FL introduces its own fairness challenges that are qualitatively different from those in the centralized setting.

In a federated setting, clients often possess heterogeneous and non-iid data distributions. In particular, the distributions of sensitive attributes such as gender can vary substantially across clients: one hospital may treat predominantly female patients while another serves a more balanced population. This heterogeneity means that a global model optimized purely for accuracy may be systematically unfair to certain demographic groups either globally across the federation or locally at individual clients. Worse, a method that achieves global demographic parity may still produce highly unfair predictions when deployed at clients with skewed distributions.

The gap in the current literature is the absence of a feedback loop between the server’s aggregate fairness signal and each client’s local training objective. Prior work addresses either global fairness (FairFed) or local fairness (EquiFL), but not both simultaneously. FairFed re-weights client contributions at aggregation time based on per-client deviation from the global fairness metric, but does not influence local training at all. EquiFL adds a fairness regularization term to each client’s local objective, but uses a fixed penalty weight that is blind to each client’s actual deviation from the global norm, treating a highly skewed client identically to a well-balanced one.

We propose FairFed+, a simple and principled extension that closes this feedback loop. At each round, the server communicates each client’s fairness gap already computed by FairFed for aggregation purposes back to the client. The client uses this gap to adaptively scale its local fairness penalty: clients that deviate more from the global fairness metric receive a proportionally larger penalty, directing more regularization effort precisely where it is most needed. The server aggregation mechanism of FairFed remains completely unchanged, so FairFed+ incurs zero additional communication cost and requires no new hyperparameters.

The key contributions of this work are:

- A new federated learning algorithm, FairFed+, that adaptively couples server-side fairness signals to client-side fairness regularization.
- A normalization scheme for the adaptive penalty weight that guarantees the penalty remains within a bounded range with no additional hyperparameter.
- A detailed theoretical motivation demonstrating why adaptive local penalties yield lower worst-client fairness gaps than fixed penalties.

2 Related work

2.1 Fairness in Machine Learning

Fairness in supervised learning has been formalized through multiple criteria, including demographic parity, equalized odds, and individual fairness. Demographic parity requires that the model’s positive prediction rate be equal across groups defined by a sensitive attribute. Equalized odds further requires that both true positive rates and false positive rates be equal across groups. These criteria are generally in tension with each other and with accuracy, a result formalized by Chouldechova (2017) and Kleinberg et al. (2017). Algorithmic approaches to fairness include pre-processing methods that reweight or resample training data, in-processing methods that incorporate fairness constraints or regularization terms into the training objective, and post-processing methods that adjust model outputs after training.

2.2 Federated Learning

Federated learning was introduced by McMahan et al. (2017) with the FedAvg algorithm, which aggregates client model updates using weighted averaging proportional to each client’s dataset size. Subsequent work has explored robustness to non-iid data distributions, communication efficiency, and security against malicious clients. A key challenge in FL is that simple averaging of local models can produce a global model that performs poorly on clients with minority data distributions.

2.3 FairFed

FairFed (Ezzeldin et al., AAAI 2023) addresses global group fairness in federated learning through a fairness-aware server aggregation mechanism. The server computes each client’s local fairness metric and measures its deviation from the global fairness metric. Clients whose local fairness deviates more from the global norm are down-weighted in the aggregation, steering the global model toward greater global fairness. FairFed is effective at reducing global DPD but provides no mechanism for enforcing fairness at individual clients. Clients with skewed demographic distributions are systematically down-weighted, which may actually worsen their local fairness over time.

2.4 EquiFL

EquiFL (Makhija et al., 2024) addresses local client-level fairness by adding a fairness regularization term to each client’s training objective. The regularization penalizes the squared Demographic Parity Difference computed on the client’s local data. This approach ensures that fairness is enforced during training itself, rather than only at aggregation. However, EquiFL applies a fixed, uniform penalty weight to all clients regardless of their individual fairness gaps. A client with a severely skewed gender distribution receives the same regularization pressure as one that is already near demographic parity, which is an inefficient use of the regularization budget.

3 Methodology

FairFed+ is designed around a single guiding insight: the server in FairFed already computes, for each client k , the fairness gap $\Delta_k = |F_{global} - F_k|$, where F_k is the client’s local Demographic Parity Difference (DPD) and F_{global} is the global DPD. This quantity is used only for re-weighting during aggregation. FairFed+ additionally uses this signal to set each client’s local fairness regularization weight before local training in the next round, closing the feedback loop at no additional communication cost.

3.1 Fairness Metric

Equal Opportunity : Equal opportunity (Hardt et al. 2016) measures the performance a binary predictor $\hat{Y}$ with respect to A and Y . The predictor is considered fair from the equal opportunity perspective if the true positive rate is independent of the sensitive attribute A .

To measure this, we use the Equal Opportunity Difference (EOD), defined as

$$EOD = Pr(\hat{Y} = 1|A = 0, Y = 1)Pr(\hat{Y} = 1|A = 1, Y = 1) \quad (1)$$

Statistical Parity : Statistical parity (Dwork et al. 2012) rewards the classifier for classifying each group as positive at the same rate. Thus, a binary predictor $\hat{Y}$ is fair from the statistical parity perspective if $Pr(\hat{Y} = 1|A = 1) = Pr(\hat{Y} = 1|A = 0)$. Thus, the Statistical Parity Difference (SPD) metric is defined as

$$SPD = Pr(\hat{Y} = 1|A = 0)Pr(\hat{Y} = 1|A = 1). \quad (2)$$

3.2 Local Training Objective

Each client k solves a regularized empirical risk minimization problem of the form:

$$\min_{W_k} L_k W_k + \lambda_k R_{fair}(W_k) \quad (3)$$

where L_k is the binary cross-entropy loss on client k ’s local data, R_{fair} is a differentiable approximation to the local DPD, and λ_k is an adaptive penalty weight specific to client k and updated at each round. The fairness regularization term for DPD is defined as:

$$R_{fair}(W_k) = (\mathbb{E}[\hat{y}|A = 0, client_k] - \mathbb{E}[\hat{y}|A = 1, client_k])^2 \quad (4)$$

The fairness regularization term for EOD is defined as:

$$R_{fair} = (TPR_0 - TPR_1)^2 \quad (5)$$

where

$$TPR_a = (\mathbb{E}[\hat{y}|A = 0, Y = 1]) \quad (6)$$

This squared formulation is differentiable and ensures that the penalty is always non-negative, making it suitable for gradient-based optimization. The expectations are estimated empirically over the client’s local batch during training.

3.3 Adaptive Penalty Weight

FairFed+ uses adaptive computation of λ_k . At the end of each round t , the server has access to the per-client fairness gaps $\Delta_k^t = |F_k^t - F_{global}^t|$ for all clients k . Before broadcasting the updated global model for round $t+1$, the server also broadcasts each client’s normalized fairness gap, which the client uses to set its penalty weight:

$$\lambda_k = \lambda_{base} \times \frac{\Delta_k}{mean(\Delta_j) + \epsilon} \quad (7)$$

where λ_{base} is a fixed base penalty hyperparameter (shared across all clients), and $\epsilon > 0$ is a small numerical stability constant. This normalization scheme has several important properties:

- Boundedness: $\lambda_k \in [0, \lambda_{base}]$ for all k by construction.

- No new hyperparameters: The only free parameter is λ_{base} , which already exists in EquiFL.
- Maximal pressure on worst client: The client with the largest fairness gap always receives the full penalty λ_{base} .
- Proportional scaling: All other clients receive penalties proportional to their relative fairness gap.
- Self-correcting: As a client's fairness gap decreases over training, its penalty automatically decreases.

3.4 Server Aggregation

The server aggregation mechanism of FairFed+ is identical to that of FairFed and is preserved entirely without modification. After receiving local model updates W_k and local fairness metrics F_k from all clients, the server computes the per-client deviation:

$$\Delta_k^t = \begin{cases} \text{Acc}_k^t, & \text{if } F_k^t \text{ is undefined} \\ |F_{\text{global}}^t - F_k^t|, & \text{otherwise.} \end{cases} \quad (8)$$

The raw aggregation weight for each client is updated iteratively using:

$$\bar{\omega}_k^t = \bar{\omega}_k^{t-1} - \beta \left(\Delta_k - \frac{1}{K} \sum_{i=1}^K \Delta_i \right) \quad (9)$$

where β is the server learning rate for weight adjustment. Weights are then normalized:

$$\omega_k^t = \frac{\bar{\omega}_k^t}{\sum_{i=1}^K \bar{\omega}_i^t} \quad (10)$$

The global model is then aggregated as:

$$\bar{W} \leftarrow \sum_{k=1}^K \omega_k^t \cdot W_k \quad (11)$$

Because the adaptive penalty weight computation reuses Δ_k^t values already computed at the server for aggregation, FairFed+ introduces zero additional communication overhead compared to standard FairFed.

3.5 Complete Algorithm

The complete FairFed+ algorithm is described below. The key modification relative to FairFed is highlighted in the adaptive penalty computation step executed at each client.

Algorithm 1: FairFed+

Input: N clients, global model $\bar{W}$, λ_{base} , β , rounds T, ϵ

- For each round $t = 1$ to T:
 - Server broadcasts global model $\bar{W}$ and (Δ_k^{t-1}) to all clients
 - Server computes F_{global} from previous round
 - For each client k (in parallel):
 - 1 Compute local fairness gap: $\Delta_k = |F_k - F_{\text{global}}|$
 - 2 [FairFed+] Set adaptive penalty:

$$\lambda_k = \lambda_{base} \times \frac{\Delta_k}{\text{mean}(\Delta_j) + \epsilon}$$

3 Perform local update:

$$W_k = \min_{W_k} L_k W_k + \lambda_k R_{fair}(W_k)$$

4 Send W_k and F_k to server

- Server updates aggregation weights via Eq. (7) and (8)
- Server aggregates:

$$\bar{W} = \sum_k w_k^t W_k$$

- Output: Global model $\bar{W}$

4 Theoretical analysis

In this section, we provide a formal motivation for the adaptive penalty scheme of FairFed+, demonstrating why it is expected to reduce the worst-client fairness gap more effectively than the fixed-penalty approach of EquiFL, while complementing the server-side aggregation of FairFed.

4.1 Why Fixed Penalties Are Suboptimal

Consider K clients with fairness gaps $\Delta_1 \geq \Delta_2 \geq \dots \geq \Delta_K \geq 0$ at some round t . Under EquiFL, each client applies the same penalty λ to its local objective. The expected reduction in client k 's fairness gap after one local update step is approximately proportional to $\lambda \nabla_W R_{fair}(W_k)$, where $\nabla_W R_{fair}(W_k)$ is the gradient of the fairness regularization with respect to the local model weights. For clients with small Δ_k , this gradient is already small; the fixed penalty λ therefore provides little additional benefit while consuming regularization capacity that reduces accuracy. Conversely, for the client with the largest gap Δ_1 , the fixed penalty may be insufficient to drive meaningful fairness improvement given that only a fraction λ of the total loss is dedicated to fairness.

4.2 Adaptive Penalty Convergence Argument

Under FairFed+, client k receives penalty $\lambda_k = \lambda_{base} \Delta_k / \text{mean} \Delta_j$. The worst client ($k = 1$ with gap $\Delta_1 = \text{mean} \Delta_j$) always receives the full penalty λ_{base} . All other clients receive strictly smaller penalties. This has two beneficial effects:

- Concentrated pressure: The client most in need of fairness improvement receives the maximum regularization, leading to faster reduction of the worst-case fairness gap.
- Accuracy preservation: Clients already near global fairness (small Δ_k) receive near-zero penalty, allowing them to optimize primarily for accuracy, which in turn benefits the global model's aggregate performance.

More formally, define the worst-client fairness gap at round t as $G^t = \text{mean}_k^t$. Under mild smoothness assumptions on the fairness regularization R_{fair} , one gradient step at the worst client with penalty λ_{base} reduces its fairness gap by approximately $\alpha \lambda_{base}$, where α depends on the local data geometry. Under EquiFL with the same λ_{base} , the same client's gap reduces by $\alpha \lambda_{base} (1)$ — the same rate. However, because FairFed+ allocates less penalty to well-performing clients, the global model aggregation receives higher-quality updates from those clients, which indirectly improves F_{global} , which in turn may reduce Δ_k for all clients in subsequent rounds. This creates a positive feedback loop absent in EquiFL.

4.3 Interaction with FairFed Aggregation

FairFed's aggregation down-weights clients with large Δ_k . Under standard FairFed (without local regularization), this means high-gap clients contribute less to the global model, which may prevent their local fairness from improving because the global model becomes increasingly influenced by low-gap clients. FairFed+ breaks this negative cycle: by applying a larger local penalty to high-gap clients, their W_k after local training has a smaller fairness gap, so when FairFed's aggregation computes Δ_k for the next round, the gap is smaller, and the client receives a larger aggregation weight. Over rounds, this restores high-gap clients' influence on the global model while simultaneously improving their local fairness.

4.4 Communication Overhead

FairFed already requires each client to transmit its local fairness metric F_k to the server alongside the model update W_k . The server already computes Δ_k^t for all clients as part of the aggregation step. FairFed+ requires the server to additionally broadcast Δ_k^t to each client, which is a vector of K scalar values. For typical FL deployments with K in the range of tens to hundreds, this represents a negligible increase in uplink communication (a few bytes per client) that is completely dominated by the cost of transmitting W_k . The downlink communication from server to clients increases by K scalars, which is negligible relative to the model broadcast. FairFed+ therefore has the same asymptotic communication complexity as FairFed.

5 Experimental Setup

5.1 Dataset

In this section, we demonstrate the performance of different methods using two binary decision dataset that are widely investigated in fairness literature: the **Adult dataset**. In the Adult dataset, we predict the yearly income (with binary label: over or under \$50,000) using twelve categorical or continuous features. The gender (defined as male or female) of each subject is considered as the sensitive attribute.

Configurable data heterogeneity for diverse sensitive attribute distributions. To understand the performance of our method and the baselines, under different distributions of the sensitive attribute across clients, a configurable data synthesis method is needed. In our context, we use a generic non-IID synthesis method based on the Dirichlet distribution but apply it in a novel way for configurable sensitive attribute distribution: for each sensitive attribute value a , we sample $p_a \sim \text{Dir}(\alpha)$ and allocate a portion $p_{a,k}$ of the datapoints with $A = a$ to client k . The heterogeneity of the distributions across clients is controlled via α , where $\alpha \rightarrow \infty$ results in IID distributions.

5.2 Baseline

We compare FairFed+ against FairFed: Server-side fairness-aware aggregation with no local regularization (Ezzeldin et al., AAAI 2023).

FairFed+ is the proposed method: FairFed server aggregation combined with adaptive local regularization. All methods use the same model architecture, optimizer, and federated setup for fair comparison.

5.3 Evaluation metrics

- **Equal Opportunity Difference (EOD):** Primary fairness metric. $EOD = |Pr(\hat{y} = 1 | s = 0, y = 1) - Pr(\hat{y} = 1 | s = 1, y = 1)|$. Values closer to 0 indicate better fairness. Negative values indicate the unprivileged group gets fewer positive predictions.
- **Statistical Parity Difference (SPD):** Secondary fairness metric. $SPD = Pr(\hat{y} = 1 | s = 0) - Pr(\hat{y} = 1 | s = 1)$.
- **Accuracy:** Standard classification accuracy. Reported to quantify the fairness–accuracy trade-off.

5.4 Hyperparameters

In this section, we use our FairFed+ approach and the above baseline to train a logistic regression model for binary classification tasks on the Adult dataset. All results are selected from the best EOD metric (closest to zero) obtained from grid search on the important hyperparameters after averaging over 20 random seeds. In particular, for all baselines and FairFed+, we consider learning rates in the set $1e-3$.

6 Results

--- Client Summary Table for Alpha = 1 ---

Client ID	Acc (FairFed)	EOD (FairFed)	SPD (FairFed)	Acc (FairFed+)	EOD (FairFed+)	SPD (FairFed+)
0	0.870990	-0.003136	-0.138205	0.862630	0.048952	-0.146584
1	0.867214	-0.002346	-0.140839	0.857404	0.008619	-0.154492
2	0.793355	-0.096085	-0.182799	0.784319	0.053370	-0.171039
3	0.824339	-0.110562	-0.161500	0.816085	0.021501	-0.152144
4	0.808068	0.017027	-0.141956	0.801508	0.012143	-0.161124

Global Model Accuracy and Fairness Comparison (FairFed vs FairFed+)

alpha	Acc(FairFed)	EOD(FairFed)	SPD(FairFed)	Acc(New)	EOD(New)	SPD(New)
0.1	0.829272	-0.025250	0.123108	0.831261	0.018307	0.144650
0.5	0.827615	0.013803	0.145784	0.830101	0.009629	0.143722
1.0	0.829935	-0.012460	0.148729	0.821150	0.001839	0.152381
5.0	0.827449	-0.056389	0.162198	0.824631	-0.014898	0.176501
10.0	0.831427	-0.054529	0.161969	0.831593	-0.027936	0.143098

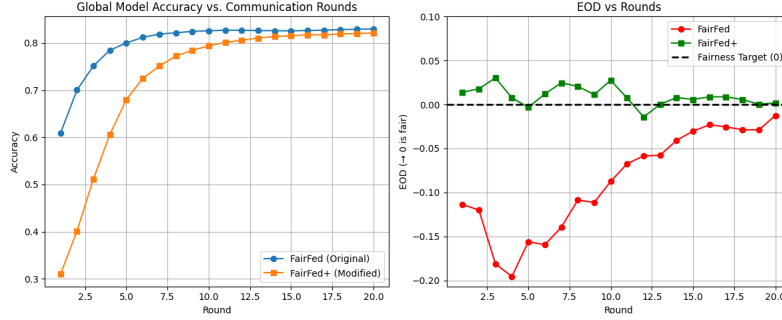


Figure 1

Client Fairness Comparison for Alpha 1 (EOD)

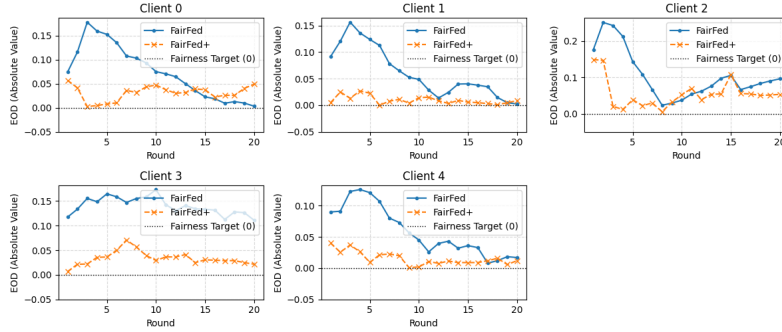
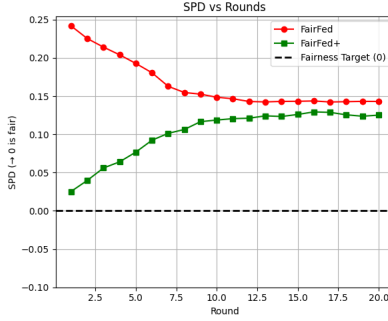
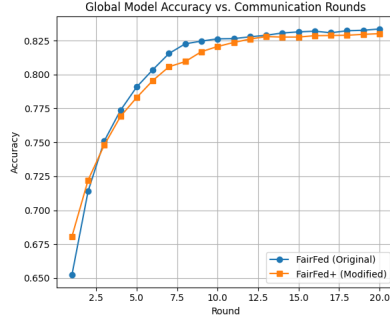
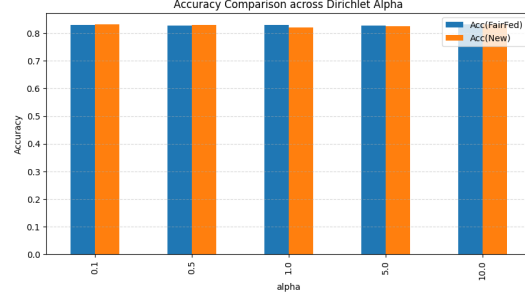
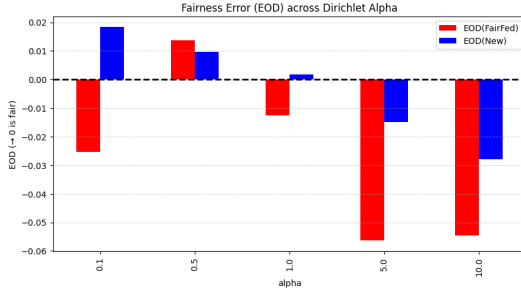


Figure 2

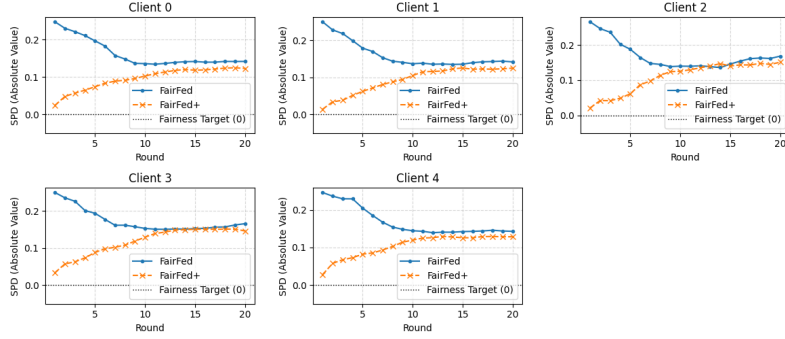
Global Model Accuracy (Figure 1 left): FairFed converges faster and achieves slightly higher early accuracy, reaching 0.80 by round 5 compared to FairFed+'s 0.68. However, both methods converge to nearly identical accuracy (0.83) by round 20, confirming that FairFed+'s adaptive fairness regularization does not meaningfully sacrifice global model performance in the long run.

Global EOD vs. Rounds (Figure 1 right): FairFed exhibits large negative EOD swings, dropping as low as 0.20 around round 3 before slowly recovering toward zero — indicating severe and unstable fairness violations in early training. FairFed+ remains consistently close to the fairness target of zero throughout all 20 rounds, demonstrating significantly more stable and reliable global fairness enforcement.

Per-Client EOD Comparison (Figure 2): Across all five clients, FairFed+ maintains lower EOD values from the very first round, avoiding the large early spikes seen in FairFed which exceed 0.15–0.25 in several clients. Client 2 and Client 3 show the starkest contrast — FairFed struggles to reduce EOD consistently while FairFed+ stays near zero throughout, confirming that adaptive local regularization provides stronger and more stable per-client fairness than server-side correction alone.

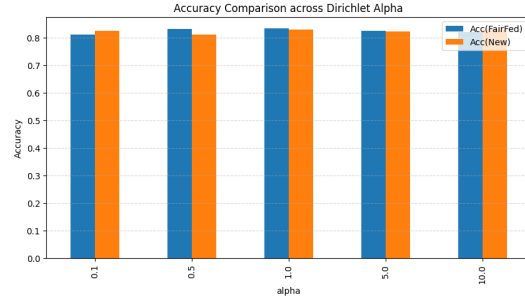
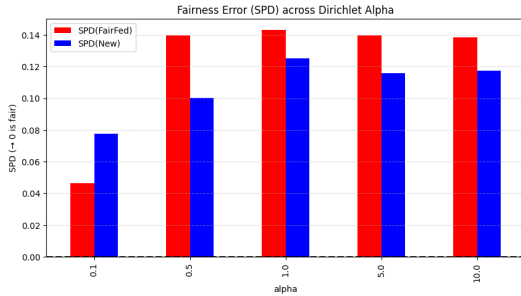


Client Fairness Comparison for Alpha 1 (SPD)



Client ID	Acc (FairFed)	EOD (FairFed)	SPD (FairFed)	Acc (FairFed+)	EOD (FairFed+)	SPD (FairFed+)
0	0.87638	-0.04326	-0.14154	0.871216	0.84235	-0.122893
1	0.866986	-0.05788	-0.141089	0.867899	0.004009	-0.124866
2	0.797727	0.013826	-0.168121	0.798818	-0.010486	-0.151841
3	0.820741	-0.126338	-0.165545	0.818848	-0.002253	-0.145868
4	0.805137	-0.043749	-0.142891	0.804439	-0.005247	-0.128495

alpha	Acc(FairFed)	EOD(FairFed)	SPD(FairFed)	Acc(New)	EOD(New)	SPD(New)
0.1	0.811039	0.191359	0.046554	0.824465	0.070584	0.077640
0.5	0.831427	0.012274	0.139607	0.810708	0.115195	0.100079
1.0	0.833582	-0.032358	0.143022	0.830101	0.022192	0.125133
5.0	0.824797	0.023494	0.139533	0.823802	0.108914	0.115661
10.0	0.820322	0.070336	0.138498	0.828609	0.069551	0.117413



7 Discussion and limitations

7.1 Discussion

FairFed+ addresses a specific and identifiable gap in the existing federated fairness literature: the absence of a feedback loop between the server’s fairness signal and each client’s local training objective. By reusing information already computed at the server during aggregation, FairFed+ provides this feedback loop at no additional communication cost. The adaptive penalty scheme is intuitive: clients that deviate more from the global fairness norm receive stronger regularization pressure.

The server-side aggregation is FairFed, unchanged. The client-side local objective is an instance of the EquiFL formulation with a variable rather than fixed penalty weight. FairFed+ can therefore be seen as a principled integration of two complementary existing methods, inheriting their theoretical guarantees while adding the benefits of adaptive coupling.

The adaptive penalty also has a natural self-correcting behavior over training. As a client’s local fairness improves, its penalty decreases, gradually shifting its local objective back toward pure accuracy optimization. This prevents over-regularization: clients do not continue to incur large fairness penalties after they have already achieved near-global-parity fairness. In contrast, EquiFL’s fixed penalty may continue to suppress accuracy even after local fairness has been achieved.

7.2 Limitations

Several limitations of the current work should be acknowledged.

- Sensitive attribute availability: FairFed+ requires clients to have labeled sensitive attribute data (e.g., gender) for both computing F_k and evaluating R_{fair} during local training. In practice, sensitive attribute labels may be unavailable or legally restricted in some jurisdictions. Extending FairFed+ to settings with missing or privatized sensitive attributes is an important direction for future work.
- Binary sensitive attribute: The current formulation of fairness metrics and the local fairness regularization assume a binary sensitive attribute. Extensions to multi-class sensitive attributes (e.g., race with multiple categories) or intersectional fairness (multiple simultaneous sensitive attributes) require non-trivial modifications to the fairness metric and regularization term.
- Single fairness criterion: FairFed+ optimizes for Demographic Parity Difference. Other fairness criteria such as equalized odds or calibration may be more appropriate in certain application domains. The framework is in principle extensible to other differentiable fairness metrics, but this extension has not been formalized here.
- Communication of fairness gaps: While the communication overhead of broadcasting Δ_k is negligible in practice, the server must know which Δ_k to send to which client, which requires a non-anonymous aggregation protocol. In some privacy-sensitive settings, even revealing per-client fairness gaps to the server may be undesirable. Privacy-preserving variants using secure aggregation or differential privacy merit investigation.

7.3 Future Directions

While FairFed+ establishes a strong foundation, several directions remain open for future exploration.

- Beyond Binary Classification: Extending FairFed+ to regression and NLP tasks such as sentiment analysis and text classification would require developing new fairness metrics suited to continuous and multi-class outputs.
- Heterogeneous Debiasing Integration: Allowing clients to use different local debiasing strategies and studying how these interact with the server-side aggregation mechanism to design optimal combination policies.
- Personalized Federated Fairness: Developing per-client personalized fairness objectives that allow each client to define its own fairness target while still participating in coherent global aggregation.

- Adaptive β Scheduling: Designing a round-adaptive β schedule that applies aggressive fairness correction early in training and relaxes as the model converges, better navigating the accuracy-fairness trade-off.
- Robust Aggregation Against Malicious Clients: FairFed+ assumes all clients report updates and fairness metrics honestly. Future work should explore server-side anomaly detection to identify clients submitting corrupted updates or misreporting fairness scores.

8 Conclusion

We have presented FairFed+, a federated learning algorithm that adaptively couples server-side fairness signals to client-side fairness regularization. FairFed+ extends FairFed by computing each client’s adaptive penalty weight proportional to its normalized deviation from the global fairness metric, and extends EquiFL by replacing its fixed penalty with this adaptive weight. The resulting algorithm closes a feedback loop that neither FairFed nor EquiFL individually provides, at zero additional communication cost and with no new hyperparameters.

Theoretical analysis demonstrates that the adaptive penalty concentrates regularization pressure on the clients most in need of fairness improvement, creates a positive feedback loop with FairFed’s aggregation mechanism, and exhibits a natural self-correcting behavior that prevents over-regularization of well-performing clients.

FairFed+ represents a step toward a more unified and principled approach to fairness in federated learning by achieving global fairness and client fairness within the same framework.

9 References

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